

PREDICTING ENERGY CONSUMPTION IN FRANCE: A VISUAL ANALYSIS OF SPATIAL RELIABILITY AND MODEL UNCERTAINTY

École Polytechnique, ENSAE

Anahí Reyes Miguel

April 17, 2026

1. Introduction: Dataset and Purpose

Accurate electricity demand forecasting is a prerequisite for modern grid stability. Over-prediction wastes generation capacity and under-prediction risks blackouts. In a country as climatically and demographically diverse as France, a model that performs well on average may still fail systematically in specific regions, which translates to producing forecasts that look acceptable in aggregate while remaining unreliable where they might matter most.

This project investigates a deceptively simple question: **is predictive accuracy uniform across geographic space?** The answer has practical consequences for grid operators, urban planners, and anyone making infrastructure decisions based on regional energy forecasts.

2. Dataset Description

The analysis combines three publicly available sources to construct a daily, region-level panel for metropolitan France covering 2017–2021:

- **Energy consumption:** RTE / data.gouv.fr, "Consommation quotidienne brute régionale"¹. Half-hourly regional electricity consumption aggregated to daily totals.

1. anahi.reyes-miguel@polytechnique.edu

1. <https://www.data.gouv.fr/datasets/consommation-quotidienne-brute-regionale>

- **Climate data:** Météo-France, "Données climatologiques de base quotidiennes"². Daily observations from 3,900+ weather stations, aggregated to regional averages.
- **Demographics:** INSEE, "Population municipale des régions"³. Annual regional population counts used to normalize consumption to a per-capita target and to derive population density.

The final modeling dataset contains 31,738 observations (region $\times$ day) across 12 metropolitan French regions (Corsica excluded). After an 80/20 train-test split, models were trained on 25,390 observations and evaluated on a held-out test set of 6,348 observations. The model performance is evaluated using three metrics: MAE (Mean Absolute Error, in kWh/capita), RMSE (Root Mean Squared Error, in kWh/capita), and MAPE (Mean Absolute Percentage Error, a unit-free ratio). RMSE penalises large errors more heavily than MAE; MAPE expresses error as a fraction of the true value, making it interpretable across regions with different consumption levels.

Table 1 classifies the key variables by data type and summarizes their role in the visualization pipeline.

Variable	Type	Role / Design Rationale
Region name	Categorical (Nominal)	Grouping, spatial position, color hue encoding
Energy consumption per capita	Quantitative Sequential	Target variable
Population density	Quantitative Sequential	Explanatory variable
MAE, RMSE, MAPE	Quantitative Sequential	Evaluation & mapping
Prediction error	Quantitative Diverging	Distribution visualization

Table 1: Data Types Classification and Visualization Role

3. Narrative Structure

The story follows a **Beginning** $\rightarrow$ **Middle** $\rightarrow$ **End** arc structured as a **Martini-glass**: it opens in author-driven mode, guiding the reader through the model benchmark and establishing that the Random Forest performs well on average ($R^2 = 0.941$) before widening into reader-driven exploration once the spatial twist is introduced.

2. <https://www.data.gouv.fr/datasets/donnees-climatologiques-de-base-quotidiennes>

3. <https://www.data.gouv.fr/datasets/population-municipale-des-regions-france-entiere>

The **beginning** presents the five-model comparison and the scatter plot of predicted vs. observed values. The **middle** is the twist: a one-way ANOVA reveals that prediction errors are spatially structured and statistically significant, with a 52% gap between the best and worst predicted regions. The choropleth map and quantile dot plot then invite the reader to explore that heterogeneity directly. The **end** explains the pattern through urban economics —building typology, the landlord-tenant problem, and hybrid work —reframing the conclusion from “the model is accurate” to “the model is *selectively* accurate.”

4. Design Rationale

Figure 1 (Viz B) Scatter Plot: Predicted vs Observed

The scatter plot maps two quantitative continuous variables to the two most accurate visual channels: horizontal and vertical position. Per Stevens’ Power Law, positional encoding is the most precise channel for quantitative comparison. A dashed 45° diagonal serves as the “perfect prediction” reference. Points are color-coded by region using a categorical palette of 12 distinct hues, and clicking any legend entry triggers a brushing interaction that highlights only that region’s points across the chart (see Section 5).

Figure 2 (Viz C) Choropleth Map: Spatial Error Distribution

The Mean MAPE, a non-negative, single-direction quantity, is encoded with a **sequential color scale** from green (low error) to dark red (high error). The sequential palette is monotonic and maps intuitively to the good/bad judgment the reader needs to make.

Figure 3 (Viz D) Quantile Dot Plot: Error Distributions

Each region’s error distribution is shown as a quantile dot plot (20 quantiles; each dot = 5th percentile). Regions are ordered by **median** MAPE rather than mean, since the median is more robust to the extreme error events visible in the tails. The mean is retained for the map and ANOVA table to remain consistent with standard reporting conventions.

Figure 4 (Viz E) Density vs MAPE Scatter

Population density spans four orders of magnitude, making a **logarithmic x-axis** the correct choice to prevent Île-de-France from collapsing all other regions into a visual cluster.

Each point represents one region, labeled directly; Île-de-France and Provence-Alpes-Côte d’Azur are explicitly annotated as the two poles of the reliability spectrum.

5. Technical Implementation

Data cleaning, feature engineering, and modeling were conducted in Python (pandas, scikit-learn, scipy). The pipeline included VIF-based multicollinearity filtering, per-capita normalization, an 80/20 train-test split, and hyperparameter tuning via GridSearchCV (5-fold CV). The Random Forest was selected for its best balance of accuracy and interpretability. Model outputs were exported as JSON files consumed directly by the visualization layer.

All interactive visualizations were built in D3.js (v7) and delivered as a single HTML page. Each chart is encapsulated in its own module (vizB.js, vizC.js, vizD.js, vizE.js, vizANOVA.js). The page implements three interaction techniques: **tooltips** on all charts (showing region name, observed/predicted values, MAE, and MAPE on hover); **brushing & linking** between the scatter plot and the choropleth map via a shared region state variable, so filtering one view filters the other; and a **sticky info panel** on the map that renders a live mini ranking of all 12 regions whenever the user hovers over a region. The choropleth boundaries use publicly available GeoJSON data.⁴

6. Findings and Conclusion

The Random Forest achieves strong average performance ($R^2 = 0.941$, $MAE = 1.706$ kWh/capita), yet a one-way ANOVA ($F = 10.03$, $p = 2.36 \times 10^{-18}$) confirms that prediction errors are not uniformly distributed across space. The gap between the best-predicted region (Provence-Alpes-Côte d’Azur, $MAPE \approx 3.4\%$) and the worst (Île-de-France, $MAPE \approx 5.1\%$) is **52%**.

One might assume that denser urban regions should be easier to predict: shared-wall collective housing is more thermally stable, producing a more learnable signal. The data proves this wrong. Île-de-France is the most densely populated region in France (1,009 people/km²) yet the least predictable, a phenomenon I call **typological blindness**. A weather-based model, calibrated on the average French home, sees a drop in temperature and predicts a sharp rise in heating demand. But well-insulated Haussmann apartment blocks require far less energy than that average.

4. <https://github.com/gregoireddavid/france-geojson>

The model has no access to building typology, so it systematically overestimates demand in dense urban zones. This is compounded by the landlord-tenant split-incentive problem (Petrov and Ryan, 2021), where the party responsible for insulation upgrades does not pay energy bills, and by hybrid work (Smite et al., 2023), which has redistributed load from predictable commercial hubs to dispersed residential circuits. As Lafourcade et al. (2021) show, the relationship between urban compactness and energy footprint is non-linear. In contrast, regions with more stable Mediterranean climates, like Provence-Alpes-Côte d’Azur, offer a more consistent signal that the Random Forest can learn with precision. This analysis shifts the focus from *predictive accuracy* to *predictive reliability across space*, a distinction that matters for decision-makers.

References

- Lafourcade, M., Carantino, B., and de Thé, C. B. (2021). The carbon ‘carprint’ of urbanization: New evidence from French cities. *Regional Science and Urban Economics*, 89.
- Petrov, I. and Ryan, L. (2021). The landlord–tenant problem and energy efficiency in the residential rental market. *Energy Policy*, 157.
- Smite, D., Moe, N. B., Hildrum, J., Gonzalez-Huerta, J., and Méndez, D. (2023). Work-from-home is here to stay: Call for flexibility in post-pandemic work policies. *Journal of Systems and Software*, 195.

A. Appendix

The interactive version of all visualizations is available at: https://anahirm.github.io/Energy_Weather_France/docs/index.html

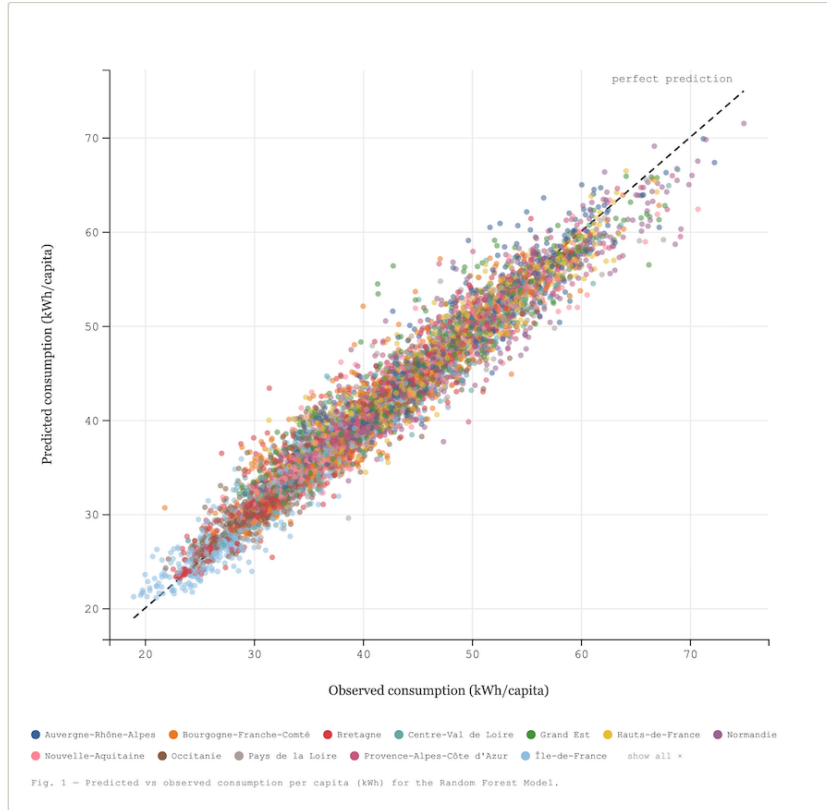


Figure 1: Viz B - Scatter plot: Random Forest predicted vs observed consumption per capita (kWh), 6,348 test observations coloured by region.

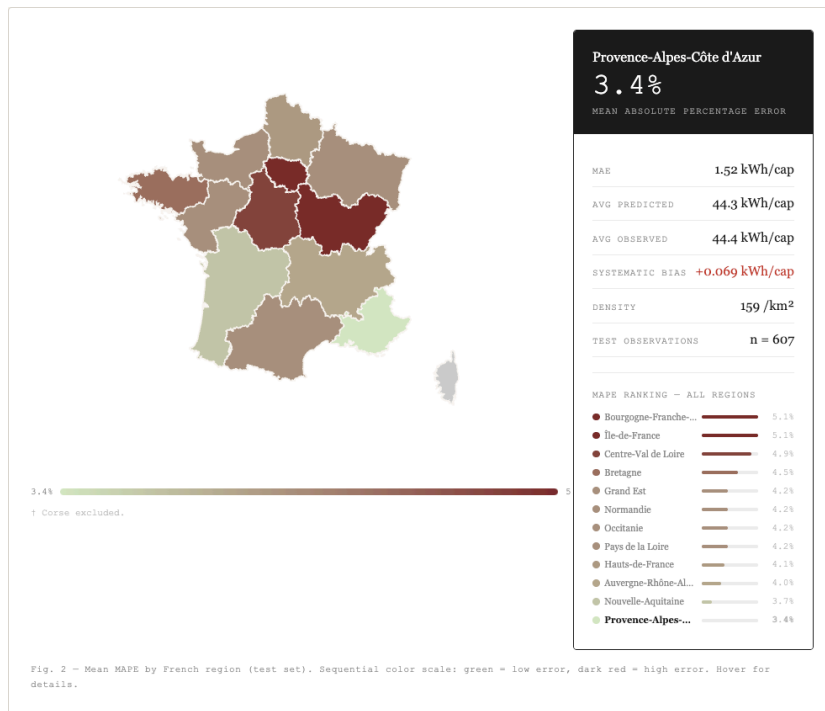


Figure 2: Viz C —Choropleth map: mean MAPE by French region. Sequential colour scale (green → dark red). Hover reveals full metric card and regional ranking.

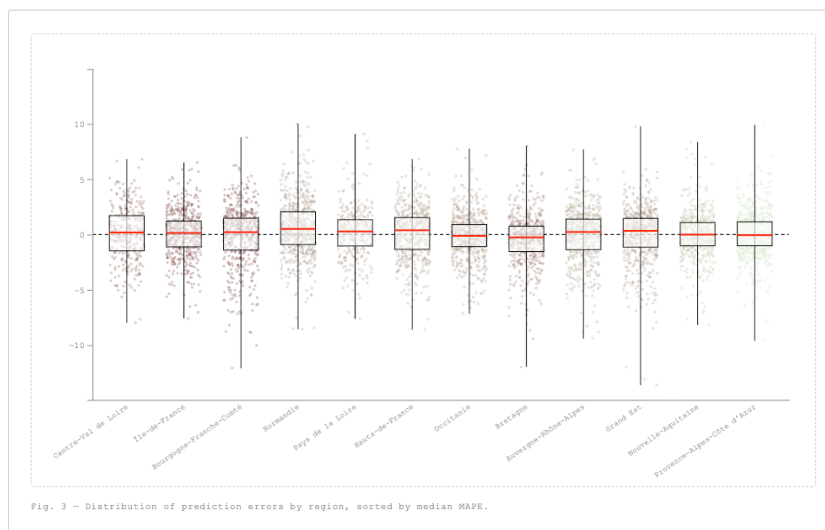


Figure 3: Viz D —Quantile dot plot: prediction error distributions by region, ordered by median MAPE (worst → best).

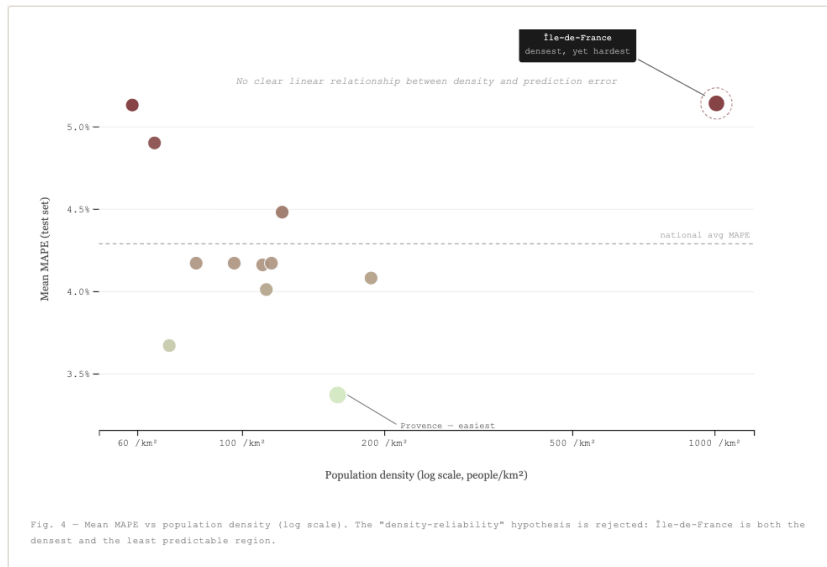


Figure 4: Viz E —Density vs MAPE Scatter. Mean MAPE vs population density (log scale). The "density-reliability" hypothesis is rejected: Île-de-France is both the densest and the least predictable region.